IT Club Election System Documentation

1. Project Overview

The IT Club Election System is a hybrid decentralized voting platform. It combines Ethereum blockchain (Sepolia) for immutable voting records with a PostgreSQL backend for high-speed data caching and IPFS for decentralized storage.

Tech Stack:

- Blockchain: Solidity, Foundry (Sepolia)
- Backend: Node.js, Express, Ethers.js
- Database: PostgreSQL
- Frontend: React, Vite, Tailwind CSS
- Storage: IPFS (Pinata)

2. Smart Contract Details (Election.sol)

The contract manages the election lifecycle through four states: Draft, Registration, Voting, and Ended. It enforces a strict ballot structure: 1 President, 1 Secretary, and 7 General Members per vote.

Core Features:

- Role-based access (Admin/Voter)
- On-chain double-voting protection
- Automatic ballot validation
- Real-time event emission for backend sync

Generated by Gemini CLI Agent for the IT Club Election Project.